STM32 Debug Board V1.0

Hardware and firmware validation report
Report revision: 1.0 · **Date:** 12 November 2026
Result: PASS — planned bench validation completed

1. Executive summary

The first revision of the custom STM32 Debug Board was successfully brought up and tested using its three debug connectors. The STM32F446ZET6 operated from its internal 16 MHz oscillator, the external 8 MHz crystal and the external-crystal-derived 180 MHz PLL configuration. SWD, JTAG, SWO trace, virtual COM port communication, user LEDs, user button, hardware reset and standalone operation were demonstrated.

No PCB design defects were identified during the planned tests, and no corrective hardware rework was required. Three practical issues were encountered and resolved or characterised: an incorrect debugger ground connection, a third-party probe's tool compatibility, and a CubeMX PLL divider setting. None required a board revision.

Outcome	Result
Planned board validation	PASS
SWD / JTAG / SWO connector coverage	3 / 2 / 3
Maximum tested CPU clock	180 MHz
Final standalone USB input current	Approximately 60 mA
Hardware defects identified / corrective rework	0 / none

This is a functional prototype validation report, not a production qualification or exhaustive MCU peripheral test. Results are based on the project owner's bench observations and debugger checks; the report does not establish that every manufactured board has been tested.

2. Hardware and test environment

Item	Description
Target MCU	STM32F446ZET6, LQFP144, Cortex-M4 with FPU, 512 KB flash
PCB	V1.0; 75 × 75 mm; two layers; ENIG
Manufacture	JLCPCB; five boards ordered, two top-side SMD assembled and three bare; order paid 31 October 2026
Power	J1 USB-C, 5 V nominal; F1 500 mA fuse; U2 AMS1117-3.3 regulator; D1 3.3 V power indicator
Oscillator	Y1 8 MHz crystal, PH0/PH1, C15/C16 30 pF
Controls	D2/PC13 and D3/PC14 active-low LEDs; S1/PC15 active-low user button; S2/NRST hardware reset
Debug ports	J2 ARM 20-pin (2.54 mm); J3 STDC14 (1.27 mm); J4 six-pin SWD (2.54 mm)
Software	STM32CubeMX, STM32CubeIDE, STM32CubeProgrammer; STM32CubeF4 V1.28.3; serial terminal
Probes	ST-LINK/V2-ISOL (V2J47S7); STLINK-V3MINIE (firmware upgraded); third-party blue 10-pin ST-LINK/V2-style probe (V2J48S7)
Instruments	Digital multimeter; USB inline current meter

3. Functional test results

PASS denotes an observed successful test; it does not imply qualification over all voltage, temperature or timing conditions.

ID	Test	Observed evidence	Result
P01	Power-up	D1 illuminated; no debugger required	PASS
P02	3.3 V rail	DMM approximately 3.29 V at regulator and debug reference pins	PASS
M01	Device identification	STM32F446xx, device ID 0x421, Rev A; 512 KB flash reported	PASS
M02	Flash and execution	Test firmware downloaded, executed and verified with genuine ST probes	PASS
I01	User LEDs	D2/D3 alternate at 500 ms intervals	PASS
I02	User button	Press/release messages; 20 ms firmware debounce	PASS
I03	Hardware reset	S2 restarts application; UART startup message observed with probe attached	PASS
U01	UART transmit	Startup message and 1 s heartbeat through J3 VCP	PASS
U02	UART receive	Individual received characters echoed at 115200 baud, 8N1	PASS
D01	SWD	Programming and breakpoint debugging on J2, J3 and J4	PASS
D02	JTAG	Programming/debugging on J2 and J3	PASS
D03	SWO	ITM stimulus port 0 messages on J2, J3 and J4	PASS
C01	HSI	16 MHz direct clock, application operating	PASS
C02	HSE	8 MHz direct clock, oscillator-ready and source status verified	PASS
C03	PLL	180 MHz, PLL/OverDrive/flash register checks verified	PASS
C04	SWV clock changes	Trace received at 8 MHz and 180 MHz on J3	PASS
R01	Final application	LED, UART, button and reset regression at 180 MHz	PASS
R02	Standalone	Repeated cold starts and reset at 180 MHz, USB-C only	PASS

Power consumption

Condition	Nominal input	Observed current	Estimated input power
Initial bring-up, HSI baseline	5 V	~20 mA	~100 mW
Final 180 MHz firmware, standalone	5 V	~60 mA	~300 mW

Power is calculated using nominal 5 V, not a simultaneous precision voltage measurement. These are board-level readings under different firmware/test conditions, not a controlled frequency-versus-current experiment. No transient, peak-load or thermal measurements were recorded.

4. Debug interface compatibility

Target connector and probe	SWD / flash / breakpoints	JTAG	SWO	Debugger reset	VCP
J2 — ST-LINK/V2-ISOL, ribbon	PASS	PASS	PASS	PASS	N/A
J3 — STLINK-V3MINIE, ribbon	PASS	PASS	PASS	PASS	PASS
J4 — ST-LINK/V2-ISOL, six wires	PASS	N/A	PASS	PASS	N/A
J4 — third-party 10-pin probe, three wires	PASS in CubeIDE	N/A	Not tested	Not tested	N/A

The third-party probe did not connect successfully using STM32CubeProgrammer; this is a tool/probe compatibility observation, not a failure of J4. JTAG and SWO were exercised separately; STDC14 does not carry NJTRST. Connector tests were performed during incremental firmware development; the final 180 MHz regression used J3 and standalone power, not every probe/clock combination.

Connection precautions

Disconnect target USB power and probe power before changing wiring. Use the target's own USB-C supply, check connector orientation and pin 1, and verify signal direction against the particular probe's manual. J4 pin 1 is the target's 3.3 V rail/reference; do not drive it from a probe power output. Do not connect standard ST-LINK/V2 pin 19 or third-party probe 3.3 V/5 V outputs to the powered target.

J4 pin	Target function	ST-LINK/V2-ISOL 20-pin header
1	3.3 V reference (VAPP)	1
2	GND	**20**
3	SWDIO / PA13	7
4	SWCLK / PA14	9
5	NRST	15
6	SWO / PB3	13

The mapping above was tested on the isolated probe. Pin 20 is also ground on the standard ST-LINK/V2. Refer to UM1075, Tables 4 and 5 [1]. For J2 and J3, the matching direct ribbon cable was used.

J3's VCP is target USART3: PD8 TX goes to J3 pin 14 (probe input); PD9 RX goes to J3 pin 13 (probe output). Signal names in UM2910 are from the target perspective [2].

5. Clock and firmware validation

Configuration	HSI baseline	Direct HSE	Final HSE + PLL
SYSCLK / HCLK	16 MHz	8 MHz	180 MHz
APB1	16 MHz	8 MHz	45 MHz
APB2	16 MHz	8 MHz	90 MHz
SWV core-clock setting	16 MHz	8 MHz	180 MHz
PLL	Disabled	Disabled	Enabled

Final PLL: HSE 8 MHz, M=4, N=180, P=2, Q=8, R=2. The VCO input is 2 MHz and VCO output 360 MHz. PLLP supplies 180 MHz, PLLQ supplies 45 MHz. AHB /1, APB1 /4, APB2 /2; timer clocks 90 and 180 MHz respectively. Voltage Scale 1, OverDrive and five flash wait states are configured. USB data and SDIO are disabled; the 45 MHz PLLQ clock is not a valid 48 MHz USB clock and is not used for USB.

Register-level evidence

The following expressions were evaluated in the debugger after `SystemClock_Config()` returned. Expected results were reported as observed. Bit definitions are from RM0390 [4].

Field	Register address / bits	HSE test	Final PLL test
SystemCoreClock	CMSIS variable	8000000	180000000
HSERDY	RCC_CR 0x40023800 [17]	1	1
PLLRDY	RCC_CR 0x40023800 [25]	—	1
SWS	RCC_CFGR 0x40023808 [3:2]	1 (HSE)	2 (PLL)
ODRDY	PWR_CSR 0x40007004 [16]	—	1
ODSWRDY	PWR_CSR 0x40007004 [17]	—	1
LATENCY	FLASH_ACR 0x40023C00 [3:0]	—	5

The generated source also confirms voltage Scale 1 and the PLL and bus prescaler values. `SystemCoreClock` is a calculated value, not an independently measured frequency; register checks plus functional tests establish correct source selection and successful operation.

Test application and launch configurations

Firmware uses HAL/SysTick timing; LEDs alternate every 500 ms, UART heartbeat and ITM text are emitted every 1 s, and the user button is polled with 20 ms debounce. UART3 is polling-based at 115200, 8N1, no flow control; it demonstrates individual character echo rather than sustained full-rate reception.

Launch file (in software/stm32_debug_board_test/)	Purpose
`stm32_debug_board_test.launch`	SWD
`stm32_debug_board_test_JTAG.launch`	JTAG
`stm32_debug_board_test_SWV.launch`	SWD + SWV; final HCLK 180 MHz; SWO limit 1 MHz

For SWV, start the launch, stop at main, open the SWV ITM Data Console, enable stimulus port 0, start trace and resume. The console displays `SWO: Running...`. Update the SWV core clock when changing firmware clock frequency. Launches use relative executable paths and do not require a specific probe serial number.

6. Issues encountered and lessons learned

6.1 Isolated ST-LINK ground pin selection

Symptom: J4 connections were intermittent or failed when using pin 4 of the ST-LINK/V2-ISOL as ground; debugger voltage indications were inconsistent (approximately 1.85–2.55 V).

Cause and correction: The standard and isolated ST-LINK/V2 models have different 20-pin assignments. UM1075 Rev 10, Table 5 [1], specifies GND on 12, 18 and 20 for the isolated probe. Pins 4, 6, 8 and 16 are unconnected; 10 and 14 serve SWIM-related purposes. Moving the ground wire to pin 20 restored reliable communication. The direct J2 ribbon connection also worked.

Residual observation: The isolated probe subsequently indicated approximately 2.96–2.97 V, while the board measured 3.29 V by DMM; the V3MINIE indicated about 3.27 V. The reason for the difference was not established. Reliable debug operation was demonstrated; no target rail defect was identified. Do not treat probe readings as a substitute for a direct voltage measurement.

6.2 Third-party Amazon ST-LINK/V2-style probe

Symptom: The blue 10-pin probe failed to connect in STM32CubeProgrammer, including after its firmware update and attempts using different connection/reset settings and reduced frequency. The exact cause was not established.

Working configuration: STM32CubeIDE connected over SWD, downloaded the application and hit breakpoints. Only three wires were connected, with the target powered separately:

Probe pin (tested unit only)	J4 pin	Signal
2	4	SWCLK
4	3	SWDIO
6	2	GND

No power or reset wires were connected. SWO, reset through the clone, and legacy ST-LINK Utility were not verified. Other look-alike probes may have different wiring or firmware. Prefer a genuine ST probe for reproducibility; verify the pinout of any alternative before connection.

6.3 CubeMX PLLQ divider locked when unused

Symptom: With USB and SDIO disabled, PLLQ was greyed out at /2. The generated code would have set a 180 MHz PLLQ output at a 360 MHz VCO frequency. DS10693 Rev 11, Table 43 [3], lists a maximum PLL48 output of 75 MHz. ST Community confirms the UI behaviour [5]. The configuration was not downloaded in this state.

Tested workaround: In CubeMX temporarily enable SDIO in one-bit mode, set PLLQ to /8, disable SDIO again, and verify the divider remains /8. Only then regenerate code. The generated `SystemClock_Config()` and `.ioc` both retained PLLQ=8, yielding 45 MHz, and the source diff showed no additional SDIO initialisation or pin assignments. No hand edits to generated clock code were required.

Lesson: Review disabled or greyed-out clock outputs and generated code against the datasheet, even when the peripheral is not used. If USB is added in a future design, provide a valid 48 MHz clock.

7. Conclusion and limitations

V1.0 meets the planned functional bench validation objectives. The final firmware starts reliably without a probe and supports the intended debug and diagnostic features. The ninth repository commit captures the tested 180 MHz configuration; the earlier HSI and direct HSE baselines remain in Git history (direct HSE commit `3dc5041`).

The following were outside the recorded test scope: BOOT0/system bootloader operation, independent oscillator accuracy and startup margin, exhaustive flash/RAM testing, long-duration stress testing, thermal/environmental qualification, EMC/ESD, USB data, unused MCU peripherals, maximum debug clock speeds and production yield. No claim is made that all boards, all probe variants or all clock/interface combinations have been validated.

The next project milestones are README and connector documentation, final portability and licensing review, and GitHub publication. The two third-party debugger photographs are intended for the README and are not embedded here.

References

1. STMicroelectronics, UM1075 — ST-LINK/V2 and ST-LINK/V2-ISOL user manual, Rev 10, Tables 4–5.
2. STMicroelectronics, UM2910 — STLINK-V3MINIE user manual, Rev 4, Table 4.
3. STMicroelectronics, DS10693 — STM32F446xC/E datasheet, Rev 11, Table 43.
4. STMicroelectronics, RM0390 — STM32F446xx reference manual, RCC, PWR and FLASH chapters (device documentation).
5. ST Community, Why is the Q factor in Clock Configuration gray in Main PLL?, 7 September 2022.
6. Project evidence: V1.0 schematic, STM32CubeMX configuration, generated source, launch configurations, Git diffs and owner-reported bench results through 12 November 2026.